FALCON's Response to Xpressbees for Supply of 10k PPH Linear Cross Belt Sorter

March 28, 2025

Offer F25-00154 Rev 0

Doc. No. - Sales/FR/PT/007/R0

Kind Attention –

Mr. Rahul Agarwal

M/S Xpressbees

Offer Ref: F25-00154-00;
Date: 28-03-2025

Subject –Techno -Commercial Offer for Linear Cross Belt Sorter project

Dear Rahul Sir,

Thank you very much for inviting Falcon to offer for the sorter project We are pleased to submit our Techno-Commercial Offer for this project.

As you will note, we have done an in-depth data analysis and evaluated various solution options best suited for your requirements.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Intra-logistics Automation Technologies and references. You will also find a description of the mapping of destinations to physical chutes.

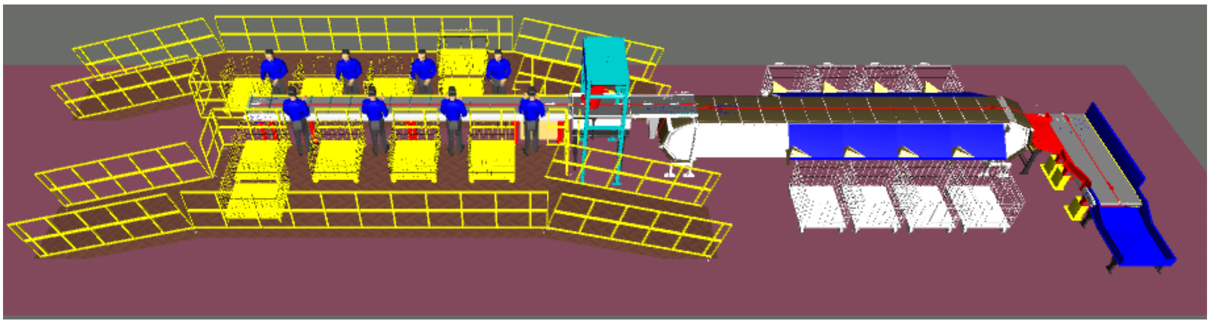
To conclude, I would like to add my personal commitment on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have and look forward to our management call to finalise.

Best Regards on behalf of the team,

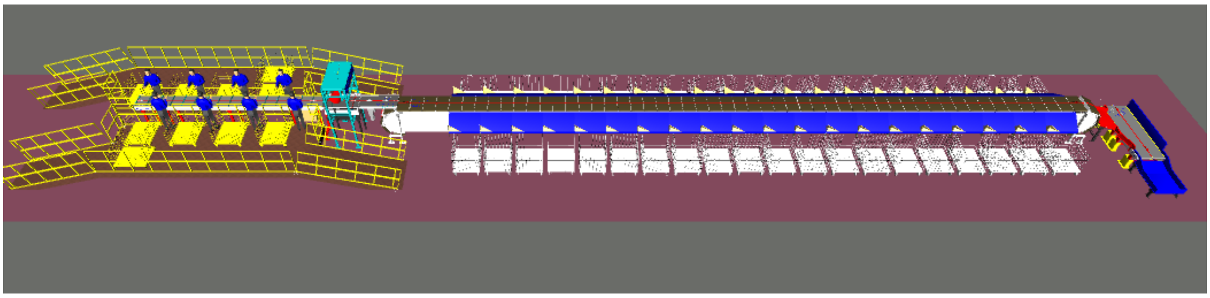
Sandeep Bansal

Chief Business Officer

Response to RFP for
Cross Belt Sorter 10k PPH



Option 1 with 8 Destination



Option 2 with 40 Destination

Proposal Reference – F24-00574 –00

Falcon Autotech Private Limited
Plot No. 87, Sector Ecotech-1, Extention-1, Greater Noida, Uttar Pradesh 201308.

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Table of Content

1. **Glossary** 6
2. **Executive Summary** 7
3. **Company Profile** 8
4. **Falcon's Experience and Achievements in Sortation Space Globally** 12
5. **Reference Projects** 15
   - 5.1 **Project 1- (CEP Client, India)** 15
   - 5.2 **Project 2- (Client – E-Commerce, India)** 16
   - 5.3 **Project 3- (Client – E-Commerce, India)** 17
   - 5.4 **Project 4- (Client – E-Commerce, India)** 18
   - 5.5 **Project 5- (CEP Client, UK)** 19
   - 5.6 **Project 6- (CEP Client, Sydney)** 20
6. **Handled Shipment Spectrum** 21
   - 6.1 **Shipment size loadable on the Sorter** 21
   - 6.2 **Shipments to be loaded on Sorter shall have the following characteristics:** 21
   - 6.3 **Shipment not loadable on the sorter** 22
7. **Proposed System Description** 23
   - 7.1 **Objective** 23
   - 7.2 **Summary of the System** 23
   - 7.3 **Main benefits of the proposed solution** 23
8. **Concept Description** 24
9. **Layout Overview** 25
10. **Proposed System Capacity Calculations** 26
11. **System description** 26
    - **11.1** Infeed Conveyor 26
    - **11.2** Scanning Conveyor 28
    - **11.3** Top Loader Conveyor 29
    - **11.4** Linear Cross Belt Sorter 29
    - **11.5** Output Chute 30
    - **11.6** Mezzanine/Operator Platform (Optional) 30
    - **11.7** Sort Fail Conveyor System (Optional) 31
12. **Description of Components of Equipment** 32
    - **12.1** Cross Belt Sorter - CBS 32
    - **12.2** CBS Carrier 32
    - **12.3** Servo Roller 33

12.4 Chassis ... 33
12.5 Linear Synchronous Motor ... 33
12.6 Power Transmission ... 34
12.7 Data Transmission ... 34
12.8 Carriers positioning System. ... 34
13. Technical specifications of CBS ... 35
14. Proposed System BOM Details ... 36
15. Electrical System ... 37
15.1 Reference Picture of Power Distribution Panel ... 37
15.2 Main Control Panel (Only for Reference) ... 37
15.3 Induct Stations Control Panel (Only for Reference) ... 38
16. Controls Architecture (Only for Ref.) ... 39
17. IT Architecture (Only for Ref.) ... 39
18. HAA Server for Sorter (Optional) ... 40
19. Falcon's WCS CONTROLIT (For Reference Only) ... 41
20. Key Components Make ... 42
21. XPRESSBEES's Responsibility ... 43
21.1 XPRESSBEES Responsibilities During the Assembly and Commissioning Phase ... 43
21.2 Responsibilities of XPRESSBEES During the Tests ... 43
21.3 XPRESSBEES Responsibilities During the Training ... 43
22. System Handover to XPRESSBEES ... 44
User Acceptance Test Parameters ... 45
23. Commercial Terms ... 47
23.1 Price Sheet ... 47
23.2 Pricing Terms: ... 47
23.3 Payment Terms ... 48
24. Warranty Period ... 48
25. Exclusions ... 49
26. Appendices ... 49

1. Glossary

S. No.	Term	Description
1	RFP	Request For Proposal
2	PPH	Shipments Per Hour
3	ICR	Intelligent Character Recognition
4	MEZZ	Mezzanine
5	FWD	Friction Wheel Drive
6	ECDS	Empty Carrier Detection System
7	AC	Alternating Current
8	DC	Direct Current
9	PLC	Programmable Logic Controller
10	IT	Information Technology
11	BOQ	Bill Of Quantity
12	I/O	Input/ Output
13	PDP	Power Distribution Panel
14	PC	Personal Computer
15	UPS	Uninterrupted Power Supply
16	CBS	Cross Belt Sorter
17	MDR	Motor Driven Roller
18	VDS	Volume Distribution System
19	NL Shipments	Non-Large Shipments
20	SL Shipments	Semi-Large Shipments
21	IPP	Individual Productivity Potential
22	DAP	Design Approval Phase
23	NO(S)	Piece(s)

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this RFQ. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success.
As prime contractor, Falcon ensures its full commitment to successfully completing this project. Following the same objective for the system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.
Our solution is based on the following key characteristics:

- **1 Sorter, based on a linear cross-belt technology, offering a design throughput of 10000 pph**
- **Single Induct Line**
- **8 Pcs/40 Pcs Destination Chute**
- **Sort Fail Chute**

A tailor-made and simple layout, specifically designed to XPRESSBEES The proposed layout, is the result of the technical requirements in the RFP document and our discussions held with the relevant stakeholders.

- Simple operational conditions due to one single Linear cross belt sorter.
- Easy maintenance: optimized number of conveyors and concentrated inducts area.

1. **Falcon's reliable and fully CE compliant Parcel sortation systems**

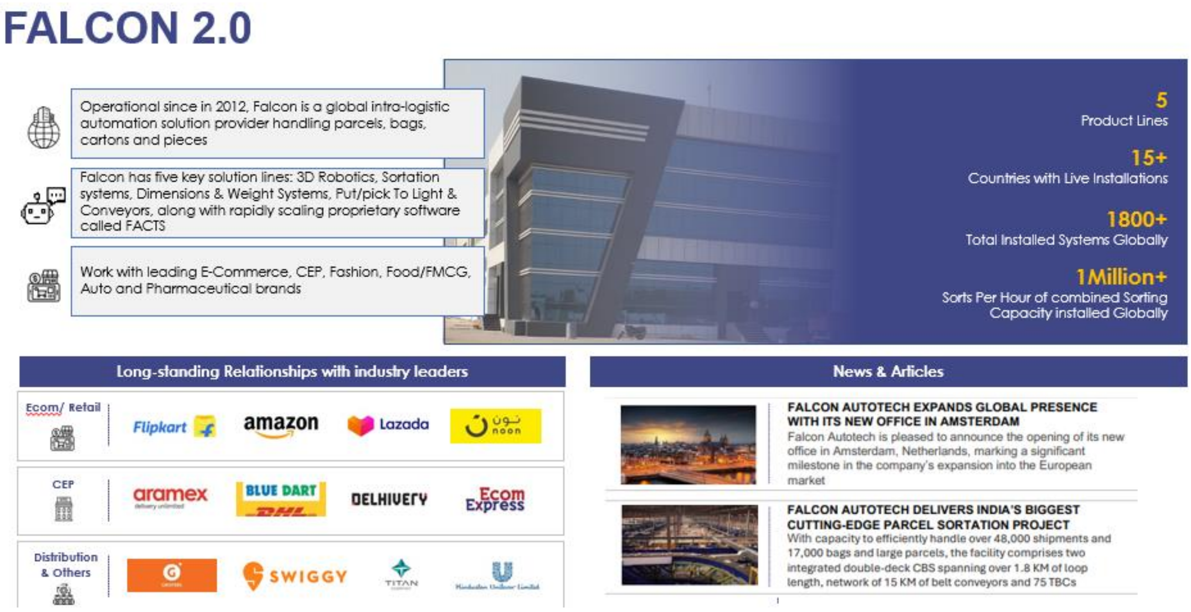
These systems are globally being used by most innovative brands such as Aramex, Amazon, Asendia, Fastway, Delhivery and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLC’s etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. **Commitment to quality systems**

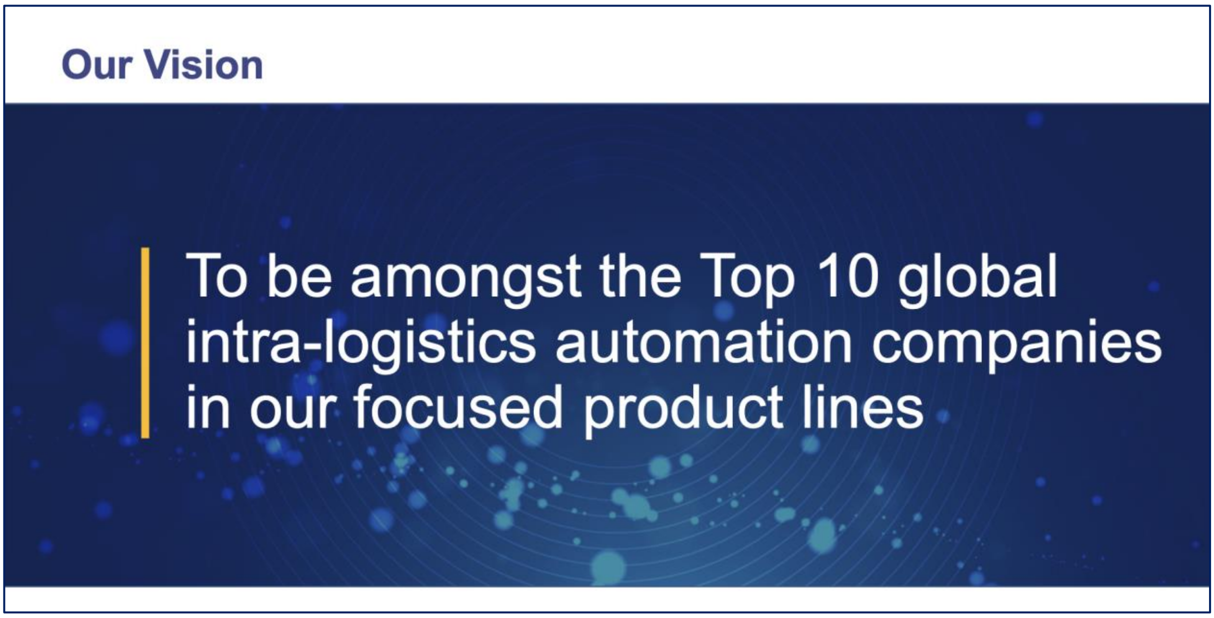
Demonstrating Falcon's clear commitment to the XPRESSBEES Group's satisfaction, the parcel sortation system, parts and services will be under warranty for 12 months from installation.

3. Company Profile

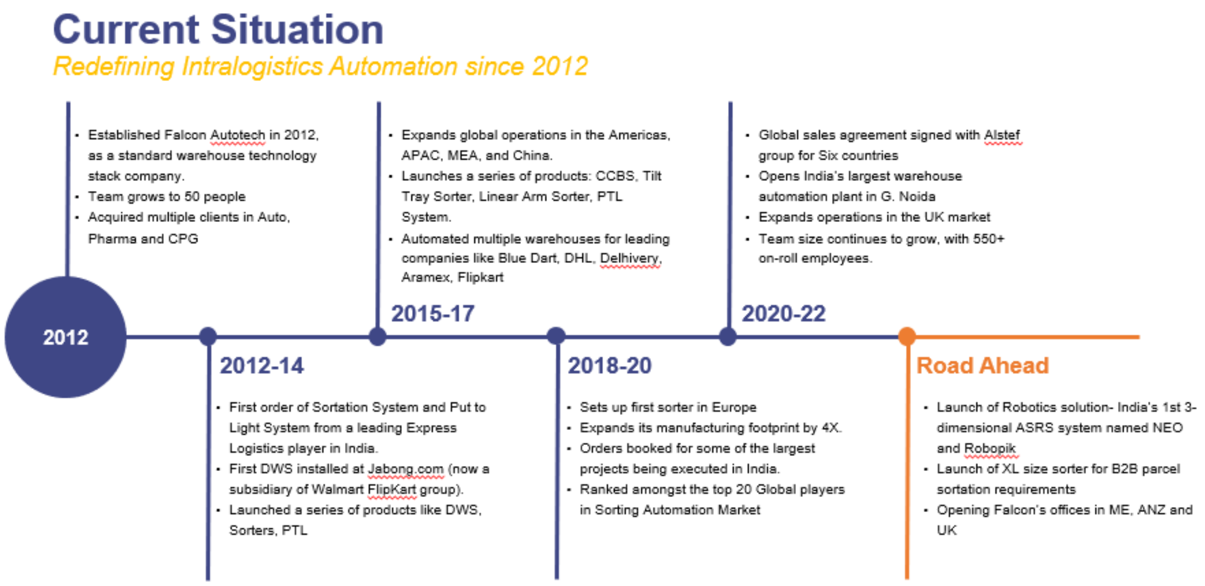
Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.



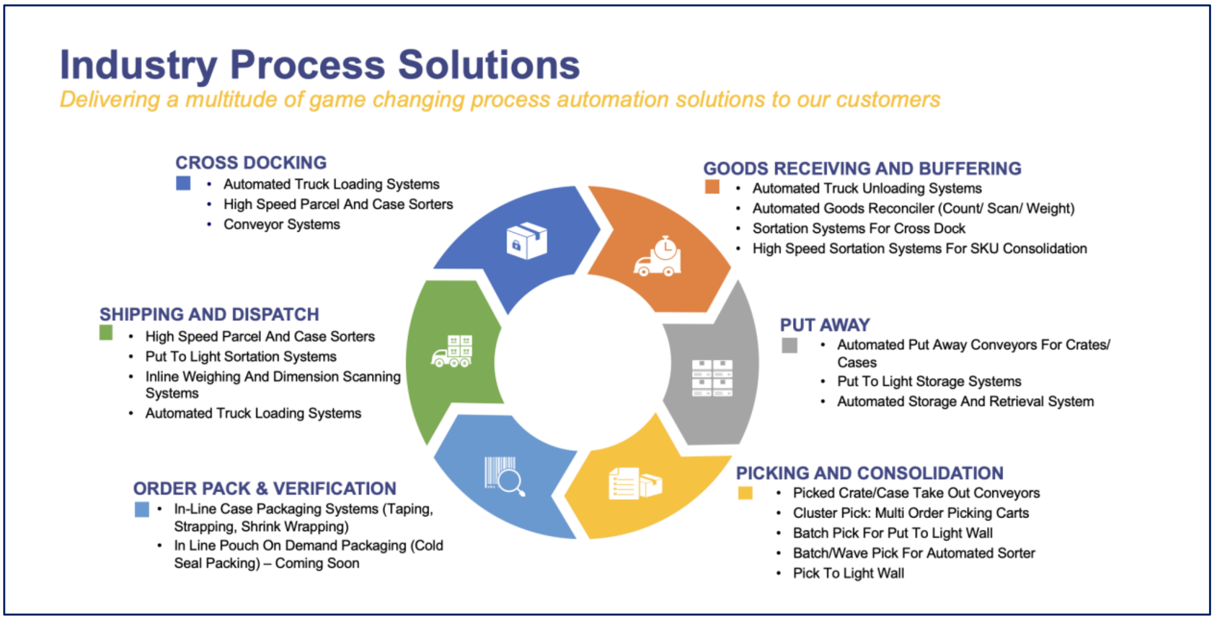
Falcon Autotech is currently among the top 15 intralogistics automation companies, our vision is to become top 10 intralogistics automation company in our focused product lines.



The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



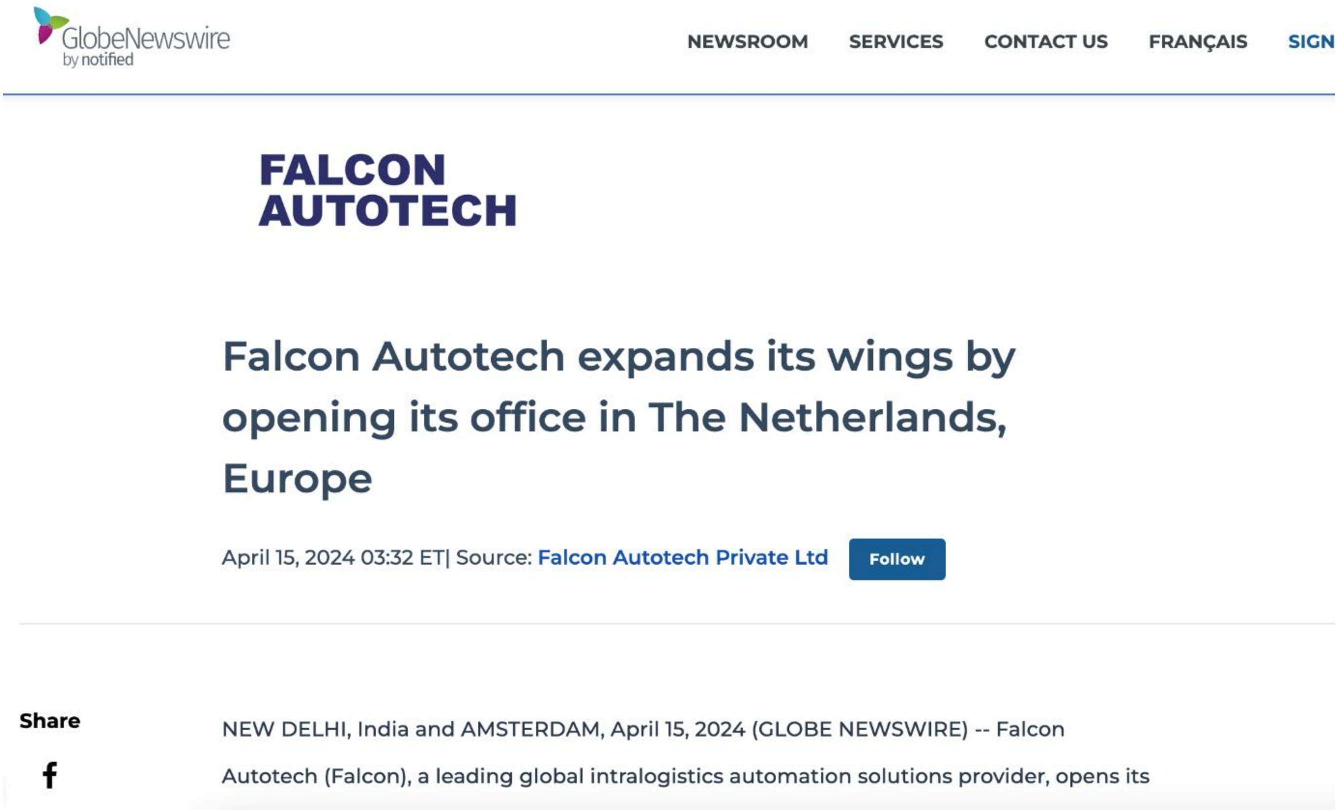
Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.

4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 15 Sortation System Suppliers** globally.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Shipments per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.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FALCON AUTOTECH

Falcon Autotech expands its wings by opening its office in The Netherlands, Europe

April 15, 2024 03:32 ET| Source: Falcon Autotech Private Ltd Follow

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NEW DELHI, India and AMSTERDAM, April 15, 2024 (GLOBE NEWSWIRE) -- Falcon Autotech (Falcon), a leading global intralogistics automation solutions provider, opens its

Breaking Barriers and Optimizing Efficiency – Journey of Delivering India's Largest Sortation System

"It was a bold decision. If it didn't work, our company would have shut down," says Naman Jain, founder and CEO of warehouse automation startup Falcon Autotech, referring to a big gamble he had taken more than a decade ago. The year was 2013, Jain and his team were negotiating a deal with a major e-commerce logistics company that had just placed an updated order for a sorter with a capacity of 6,000 parcels per hour. That was 4x the specification earlier agreed upon.

"All the founders came together, and we took a calculated risk," recalls Jain. At the time, Falcon did not have the technology to develop such sorters, and hence, the said client refused to pay any money upfront and offered to pay the machine's price in monthly installments. However, there was a caveat: if it failed even once, Falcon would simply take the machine back and refund the entire amount. Falcon's 'calculated risk' paid off. The machine worked flawlessly and was handed over to the parcel company after 36 months. "We are now doing projects worth INR100 crore. That was unthinkable 10 years back," says Jain.

Falcon Autotech has covered a lot of ground since that big bet. Last month, it installed India's largest sortation equipment at

From sorters to conveyors & robot based systems, here's the top tech warehouses are investing in

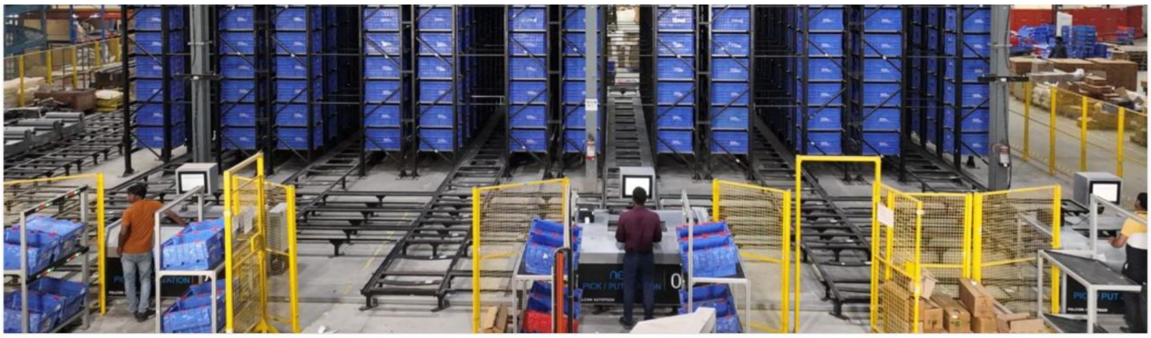
India's warehousing industry has travelled a long distance from "godowns" to modern storage facilities called Grade A warehouses.
Warehouse automation is a gradual process. India started off late, but that may actually help it bypass some of the mid-age technologies and adopt the latest ones. While several players are still warming up to automation, which are the

Falcon Announces Successful Go-Live of Automated Parcel Sortation Solution at DTDC's Chennai Facility

New Delhi, India – 2nd Aug 2023: Falcon Autotech, a leading supplier of intralogistics automation solutions, has been selected by DTDC Express Ltd, one of India's leading integrated express logistics company, to automate its parcel sorting operations at its super hub of 1,75,000 sq ft in Chennai, Tamil Nadu. Using its cross belt sorter technology, Falcon has designed DTDC's parcel sorting system, which can handle 6,000 parcels per hour, operate in a 24 X 7 environment, and can be expanded to cater to future growth.

The new linear cross-belt solution leverages cutting-edge technology to automate key sorting processes in DTDC's warehouse, including parcel profiling and sorting. This solution is designed to optimize the space requirements for sorting operations, increase efficiency, and reduce operational costs.

"We are thrilled to see our warehouse automation solution go live with DTDC, this solution is a testament to our commitment to providing innovative intra-logistics solutions that meet the evolving needs of our customers," said Falcon's CEO Naman Jain.



Transforming Indian Retail: The Impact of ASRS on Warehousing Operations

In recent years, the Indian retail industry has experienced unprecedented growth, propelling the nation to become the fifth-largest global retail destination. The industry, marked by its resilience and adaptability, went through significant transformations during the pandemic. While online shopping saw a remarkable surge, the reopening of physical stores ushered in a resurgence of the multi-sensory shopping experience. As a testament to this growth, shopping malls now encompass an astonishing 23.25 million square feet of retail space. However, the evolving consumer preferences have placed substantial pressure on businesses, necessitating the seamless integration of e-commerce and in-store experiences, which, in turn, has led to complex logistics challenges. The traditional warehousing systems struggled to cope with the demand for faster processing and the requirement for expanded storage capacities.

Historically, retailers had heavily relied on manual labor for their warehousing operations. However, the dynamism of today's retail market demands a shift towards automation. Enter Automated Storage and Retrieval Systems (ASRS), a technological

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Shipment Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focuses on Sortation Solution –

5.1 Project 1- (CEP Client, India)

The System is equipped with two fully automated and interconnected Sub-systems. Sub-System 1 is designed for handling large B2B boxes and E-commerce shipment bags while the Sub-System 2 is designed to handle Small E-commerce packages.

Solution Specifications –

- 48,000 PPH (Double Deck CBS- Shipment Sorter)
- 17,000 PPH (Double Deck CBS- Bag Sorter)
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- 2 Sets of Double Decker CBS Sorters
- Mezzanine Structures
- Automated Singulators
- Fully Automatic Inductions
- Semi-Automatic Inductions
- Telescopic belt conveyors
- PVC Belt Conveyors
- Modular Belt Conveyors
- Spiral Chutes with Braking rollers
- 5-Sided Scanning Tunnels
- High speed weighing conveyors
- Direct Bagging Chutes
- Put to Light Chutes
- Volume Distribution systems
- High Availability Server Systems
- WCS

Site Pictures –








5.2 Project 2- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C shipments. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

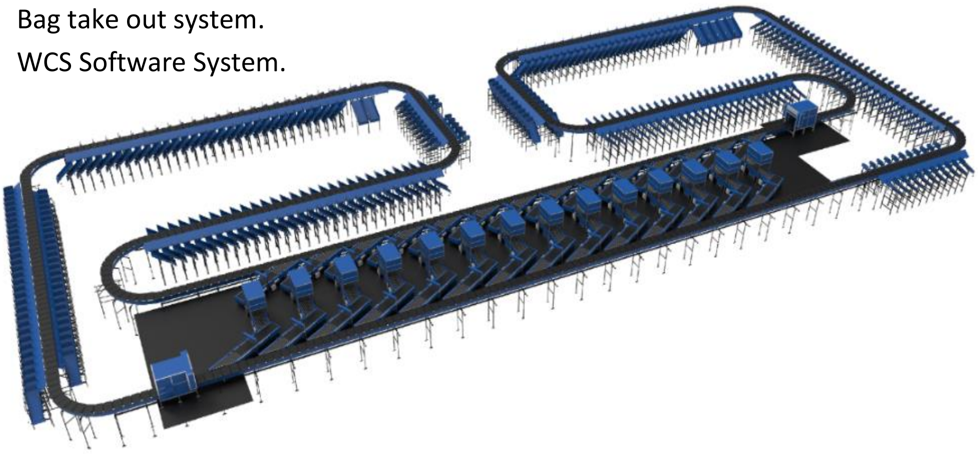
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system.
- WCS Software System.





5.3 Project 3- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Shipments.

The customer chose Falcon Autotech based on its unique design, capability of seamless integration with WMS and life cycle support services.

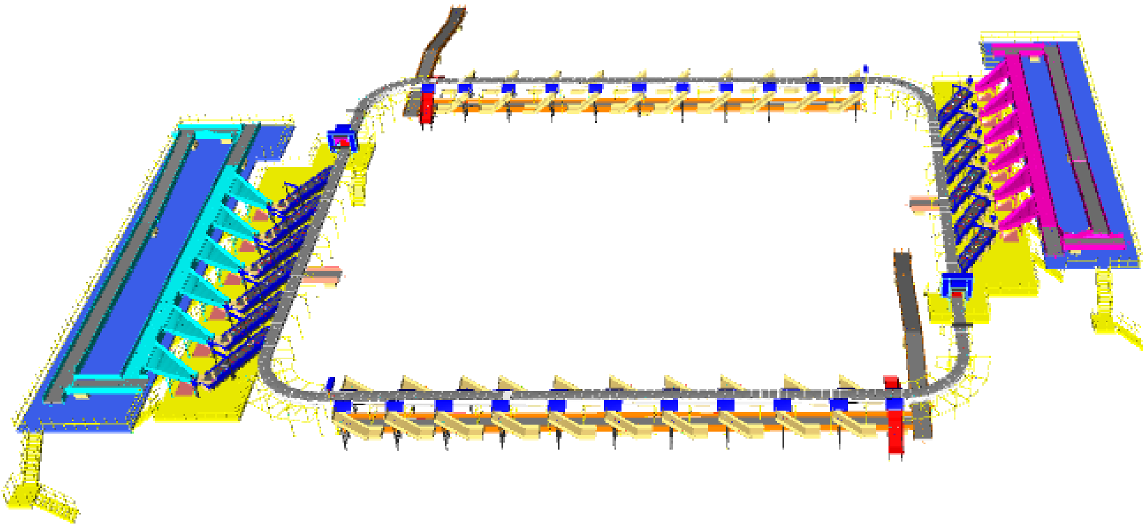
Solution Specifications –

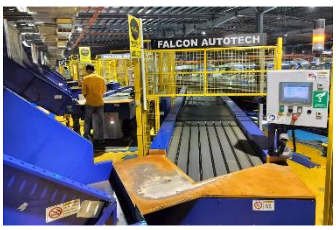
- Throughput: 24,000 PPH
- End Destinations: 40 Collection Type Chutes

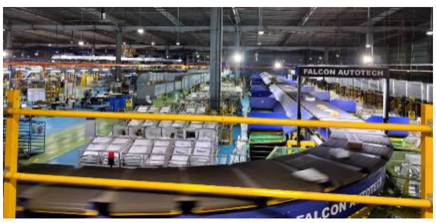
Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- Smart Collection type chutes
- Bag take out system.
- WCS Software System.

Layout and Site Pictures -







5.4 Project 4- (Client – E-Commerce, India)

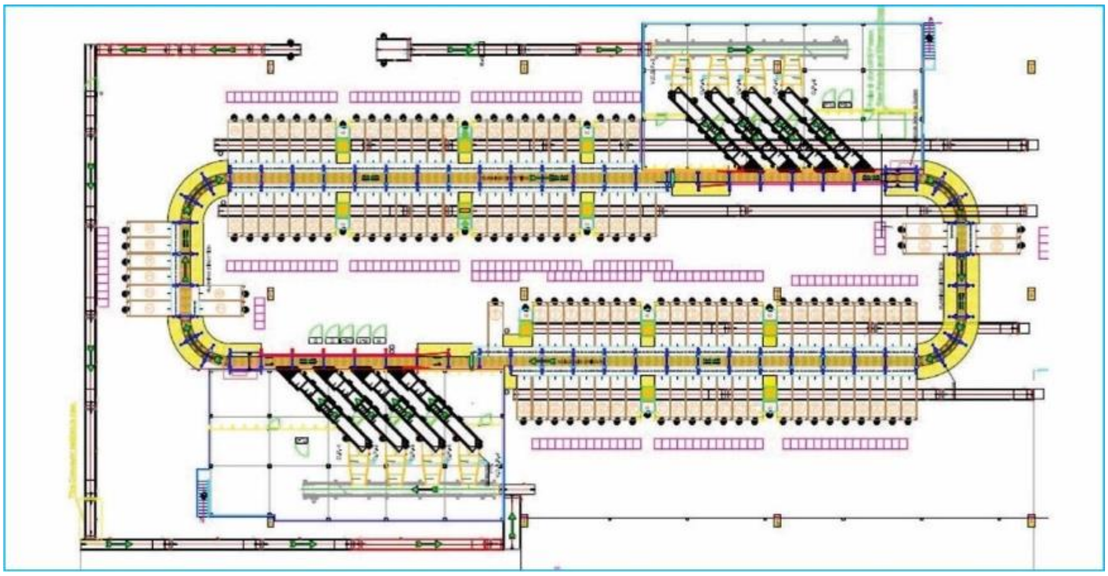
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Linear Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



5.5 Project 5- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 shipments per hour. The system is equipped with three infeed conveyors integrated with an automatic label applicator before shipments enter the sortation system. The shipments are sorted using Falcon's Linear Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Linear Cross Belt Sorter.
- WCS Software System.

Site Picture –

5.6 Project 6- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 shipments per hour with the help of Falcon's Linear Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the shipments at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight shipments.

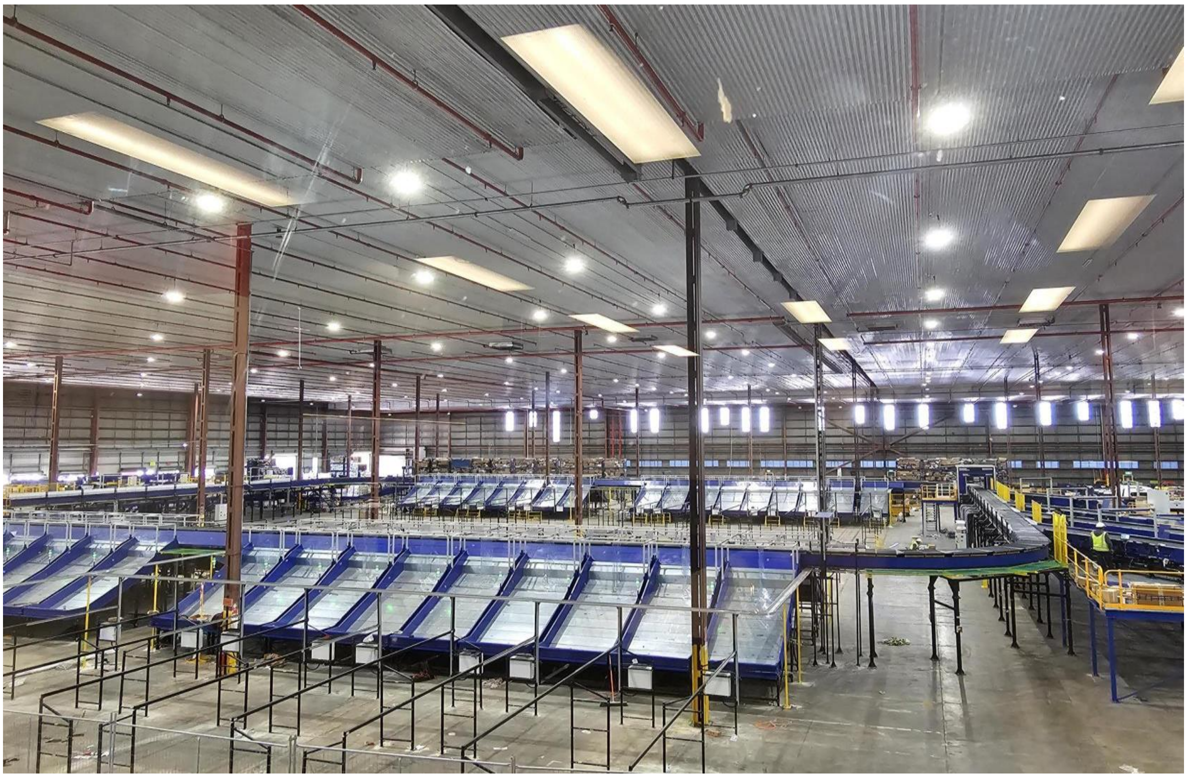
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize shipment.
- Linear Cross Belt Sorter.
- WCS Software System.

Site Picture –



6. Handled Shipment Spectrum

As per the shipment spectrum data provided in the RFP documents, Falcon has studied and analysed the shipment spectrum in detail.

Falcon proposes to use its "Linear Cross Belt Sorter technology for this solution

6.1 Shipment size loadable on the Sorter

Falcon's Linear Cross belt sorter has a capability to handle the below mentioned shipment sizes and weight.

Specification	Unit	Value
Max Length	mm	400
Max Width	mm	400
Max Height	mm	400
Max Weight	Kg	10
Min length	mm	100
Min Width	mm	100
Min Height	mm	5
Min Weight	gm	100

6.2 Shipments to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of shipment cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Shipments shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The shipments must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.3 Shipment not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

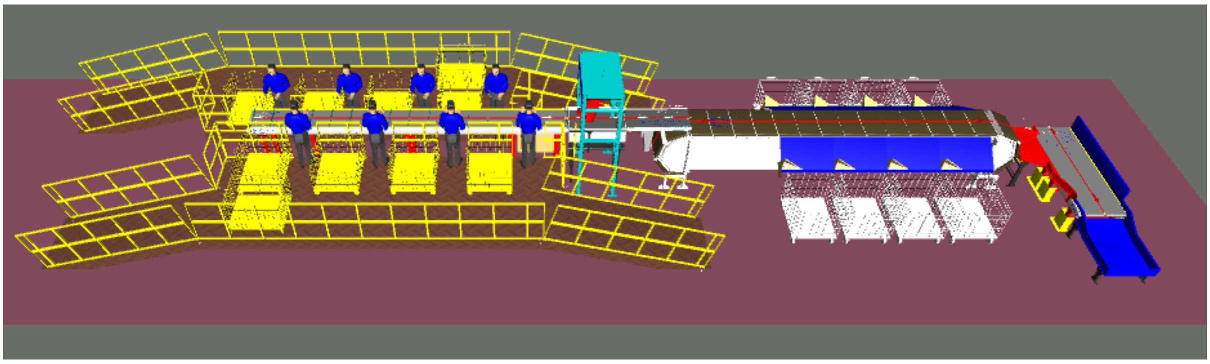
1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items.
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g., Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile shipments with contents not sufficiently secured.
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non-conveyable items.

7. Proposed System Description

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Linear CBS for sorting shipment & boxes as per XPRESSBEES Requirement

7.2 Summary of the System



Process flow:

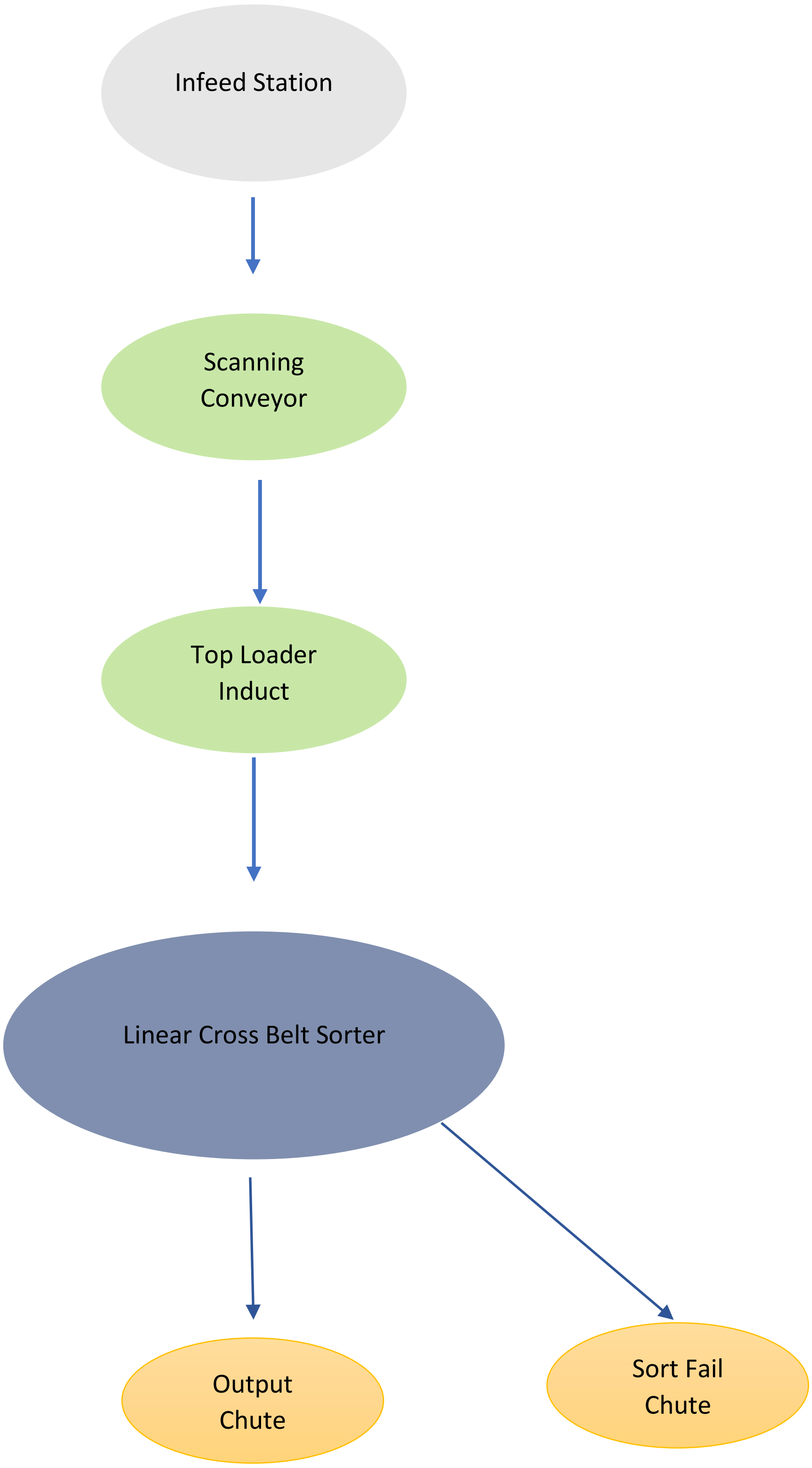
1. **Infeed System: -** Shipments are placed on the infeed lines in a singulated manner, with one parcel in each slot of the clit. The infeed conveyor features clits spaced at a pitch of 450 mm.
2. **Scanning Conveyor:** Shipments on the scanning conveyor will have their barcodes scanned using a scanner mounted above, with scanning performed exclusively from the top.
3. **Top Loader:** One top loader is planned to induct the parcels coming from scanning conveyor to the Linear Cross Belt Sorter System
4. **Linear CBS:** - Once the shipments have entered the Linear CBS, Cross-Belt Sorter (CBS) efficiently sorts the shipments into their respective output chutes by utilizing the data provided by XPRESSBEES's sorting logic.
5. **Output Chutes: -** The shipments are discharged from transfer types of chutes in the collection Bin (Client Scope).

7.3 Main benefits of the proposed solution

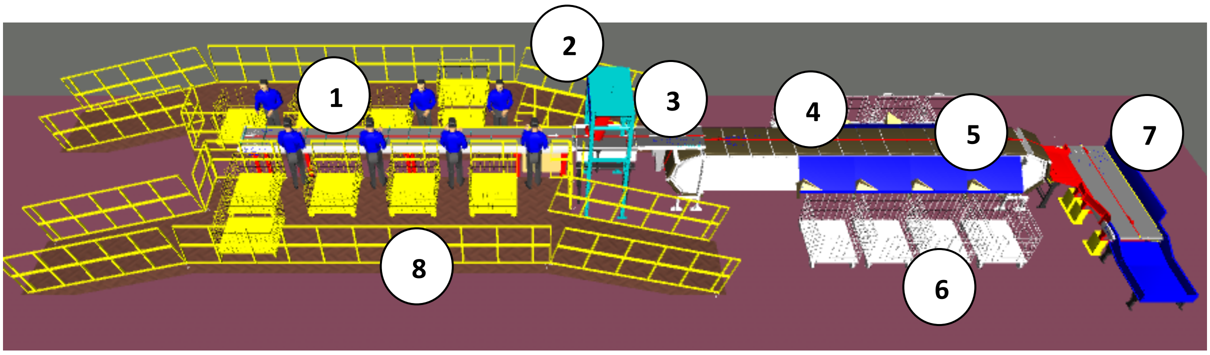
The proposed solution has the following advantages:

1. High operational throughput
2. Low occupancy of floor space in the building.
3. Narrow discharge centers for the increased number of splits in limited space.
4. FALCON's CBS can adapt to changing business requirements by adjusting its speed. to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

8. Concept Description



9. Layout Overview



Legend

1. Infeed Point
2. Scanning Conveyor
3. Top Loader Induct
4. Linear Cross-Belt Sorter (CBS)
5. Destination Chute
6. Collection Trolley
7. Sort Fail Chute
8. Mezzanine/Operator Platform

10.Proposed System Capacity Calculations

The following table shows the throughput calculation for the sortation system designed based on XPRESSBEES RFP requirements.

SPECIFICATION	VALUE
Sorter Type	Linear CBS
Speed	1.5 m/s
Dual Belt Carrier Pitch	500 mm
Carrier per hour	10,800 CPH
Belts per hour	10,800 BPH
Top Induct Feedline Capacity (Designed on avg. parcel size 400*400mm)	10,000 Shipments per hour
No of Top Induct Feedlines per System	1 Nos.

Note : Operation TPH is totally dependent on the feeding capacity of the operators and parcel size

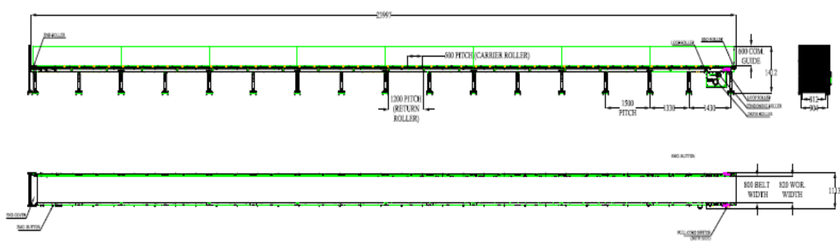
11. System description

11.1 Infeed Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.
Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.

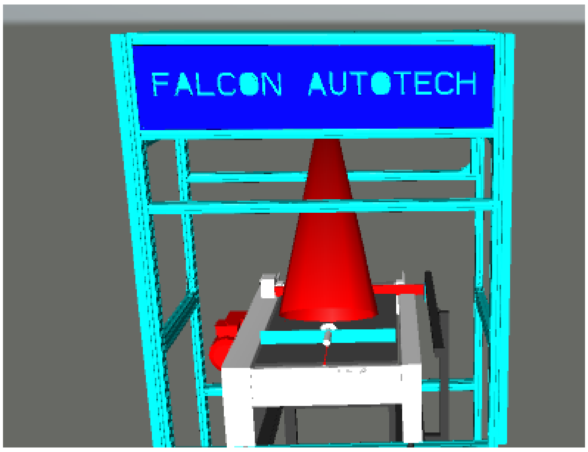


Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1200
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	30
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord/Lenze
Drive	make	Lenze/Siemens/Omron/Allen Bradly
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

11.2 Scanning Conveyor

A scanning conveyor is a conveyor system equipped with barcode scanners to automatically read shipment labels as they move along the conveyor belt. These scanners, typically mounted above the conveyor to capture barcode information without manual intervention. The scanning process helps in tracking, sorting, and routing parcels efficiently in warehouses, distribution centers, and logistics operations.

Top Sided Barcode Scanner present on Scanning Conveyor to scan 1D codes and 2D codes.

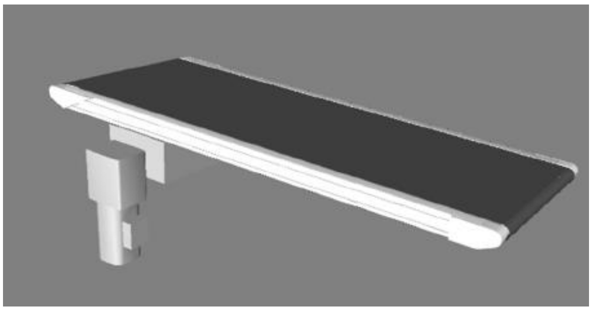


Specification	UOM	Remark
Manufacturer Name	Name	SICK/Cognex
Barcode Type	Type	1D&2D
Module Size	Mils	>=10 for 1D & >=25 for 2D
Number of Codes	Nos	1
Location	Side	Top Side Only
Orientation	Type	Omnidirectional
Color of bars	Color	Black
Under Foil	yes/no	No

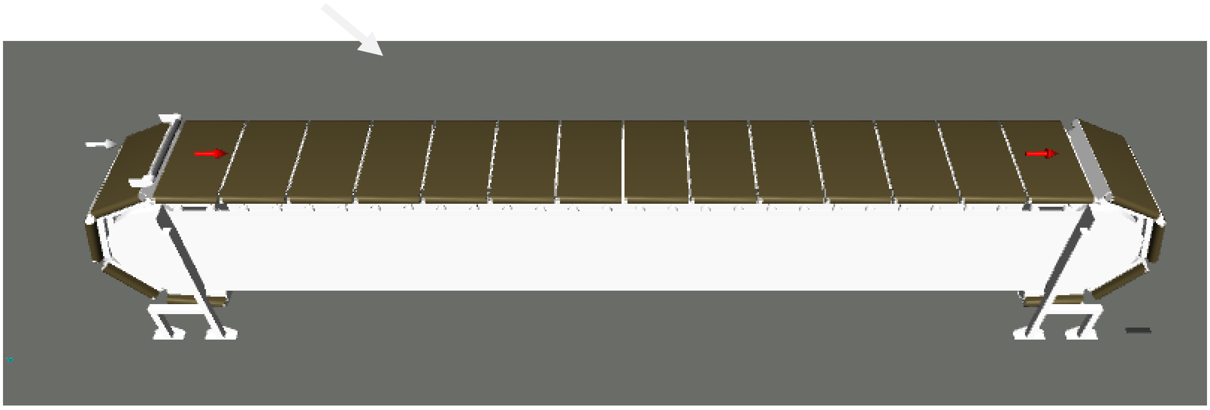
11.3 Top Loader Conveyor

A Top Loader conveyor incorporates advanced automation and control technologies to intelligently transfer stream of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this conveyor is used for inducting shipment/boxes directly on to the sorter. The Belts are PVC for smooth shipment movement.



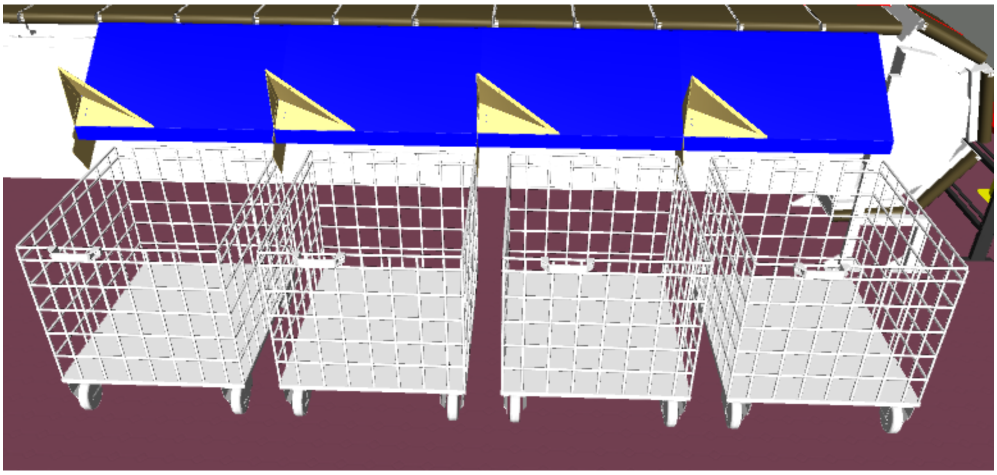
11.4 Linear Cross Belt Sorter



1. The proposed Linear CBS Sorter will be installed on Ground level. The Sorter runs at 1700 mm from Ground level.
2. We have provided Linear with Linear CBS Single Belt Carriers with 500 mm pitch and Belt Size of 495 x 800mm.

11.5 Output Chute

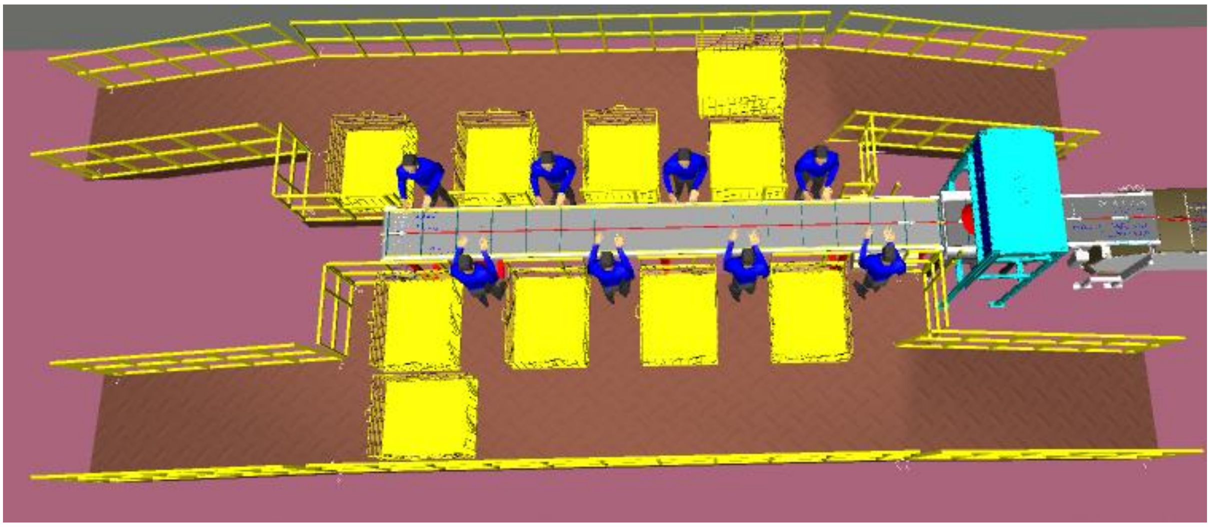
In the proposed solution the **transfer type chute is designed** for sorting directly to containers/collection trolley which will be in client scope.



11.6 Mezzanine/Operator Platform (Optional)

Two mezzanine platforms are planned near the infeed conveyor, one on each side, for the operators (as shown in the image below in brown).

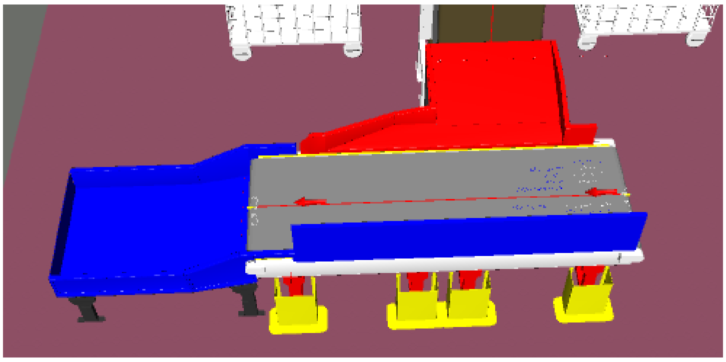
Two Ramp has been also planned each side for movement of trolleys by the operator .



Total Area: 61 Sq. Meters

11.7 Sort Fail Conveyor System (Optional)

At the end of the Linear CBS, a sort-fail conveyor system is planned to collect all parcels that fail sorting. The parcels will then be transferred to the collection chute via the conveyor system.

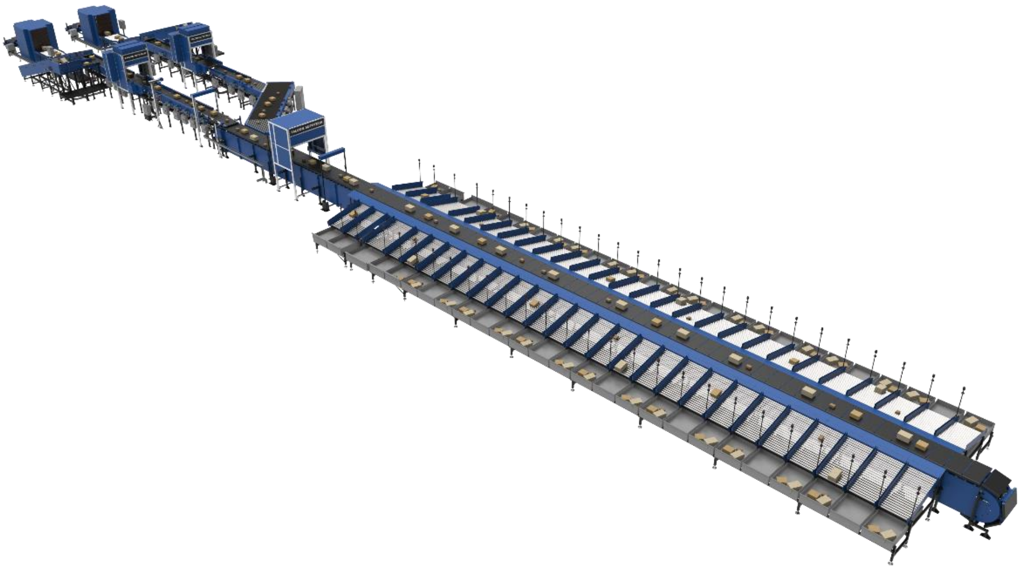


12.Description of Components of Equipment

12.1 Cross Belt Sorter - CBS

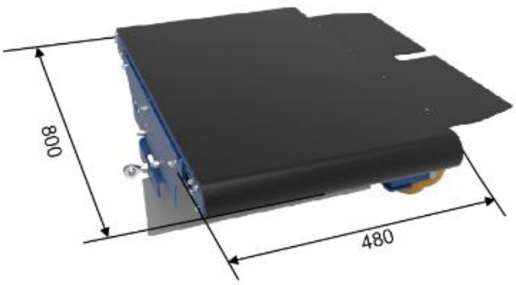
Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



12.2 CBS Carrier

Falcon's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners "and non-sortable shipment recirculation.



12.3 Servo Roller

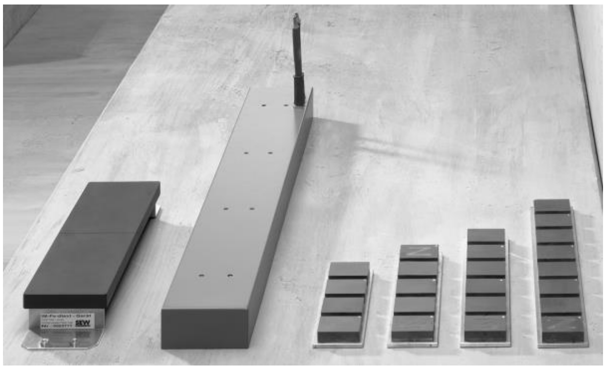
High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.

12.4 Chassis

Falcon Autotech' s Cross Belt Carrier Chassis is made up of lightweight Aluminum which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.

12.5 Linear Synchronous Motor

A Linear Synchronous Motor (LSM) is a type of motor that generates linear motion rather than rotational motion, unlike traditional rotary motors. The linear motion in an LSM is achieved by moving a magnetic field along a straight path, propelling the motor along that same path. LSMs are widely used in various applications requiring precise linear motion, such as magnetic levitation (maglev) trains, automated material handling systems, and even amusement park rides.

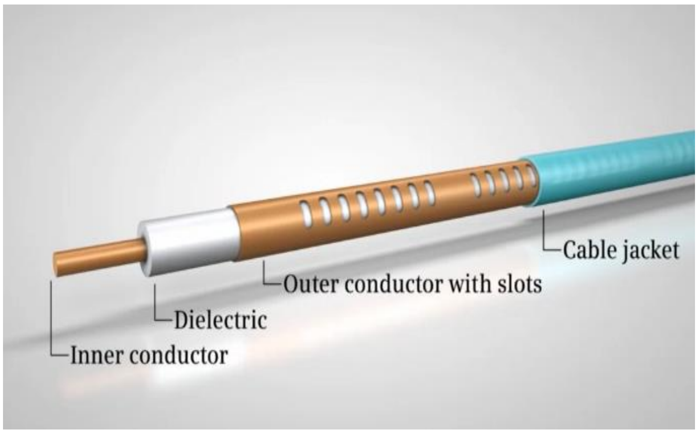


12.6 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.

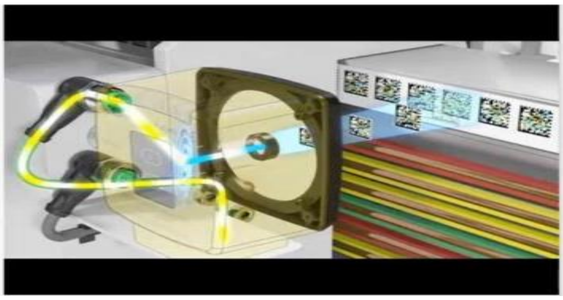
12.7 Data Transmission

The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.



12.8 Carriers positioning System.

Positioning system determines the exact location of each carrier in the Linear at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter Linear, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



13. Technical specifications of CBS

Items	Make / Number / Execution
Sorter Carrier Type	Linear CBS
Sorter Speed in m/s	1.5 m/s
Sorter Length (Top Side Only) for 8 Destination	~7 m
Sorter Length (Top Side Only) for 40 Destination	~53 m
Sorter Actuation Technology	Electric
Height of Sorter	~1700 mm
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	500
No. of Carriers for 8 destinations	35 Nos.
No. of Carriers for 40 destinations	113 Nos.
Drives	
Motor/Drive type	LSM
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

14. Proposed System BOM Details

For Option 1 with 8 Destinations:

Pos.	Qty.	Description	Value
1	1	**Infeed System** • Powered Belt Conveyors • Scanning Conveyor • Top Loader • Top Side Scanner	 ~7 metres (1 Modules) 1 No 1 No 1 Set
2	1	**Sorter Package** **1 Linear Cross Belt Sorter** • Sorter Height • Sorter Length **Including:** • Standard Sorter Supports • Emergency Stop Buttons	 1700mm ~7m
3	1	**Chute Package** • Output Chute • Sort Fail Conveyor System-Optional	 8 Nos 1 No
4	1	**Mezzanine Package (Optional)**	61 Sq. Meters

For Option 2 with 40 Destinations:

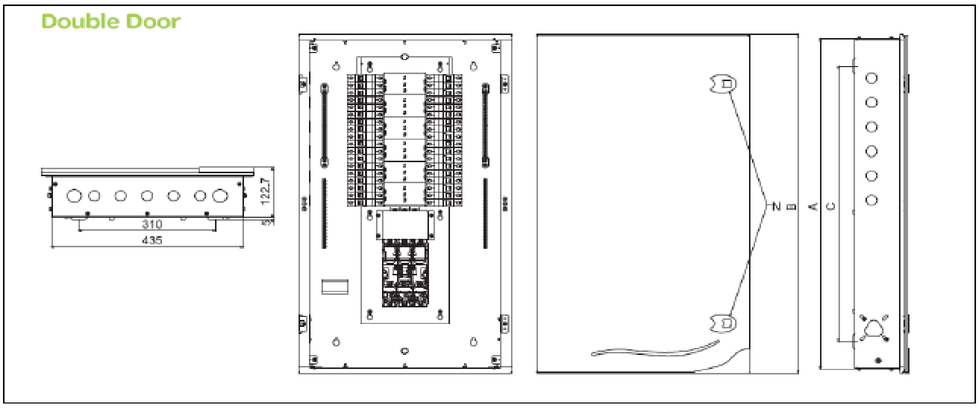
Pos.	Qty.	Description	Value
1	1	**Infeed System** • Powered Belt Conveyors • Scanning Conveyor • Top Loader • Top Side Scanner	 ~7 metres (1 Modules) 1 No 1 No 1 Set
2	1	**Sorter Package** **1 Linear Cross Belt Sorter** • Sorter Height • Sorter Length **Including:** • Standard Sorter Supports • Emergency Stop Buttons	 1700mm ~53m
3	1	**Chute Package** • Output Chute • Sort Fail Conveyor System-Optional	 40 Nos 1 No
4	1	**Mezzanine Package (Optional)**	61 Sq. Meters

15. Electrical System

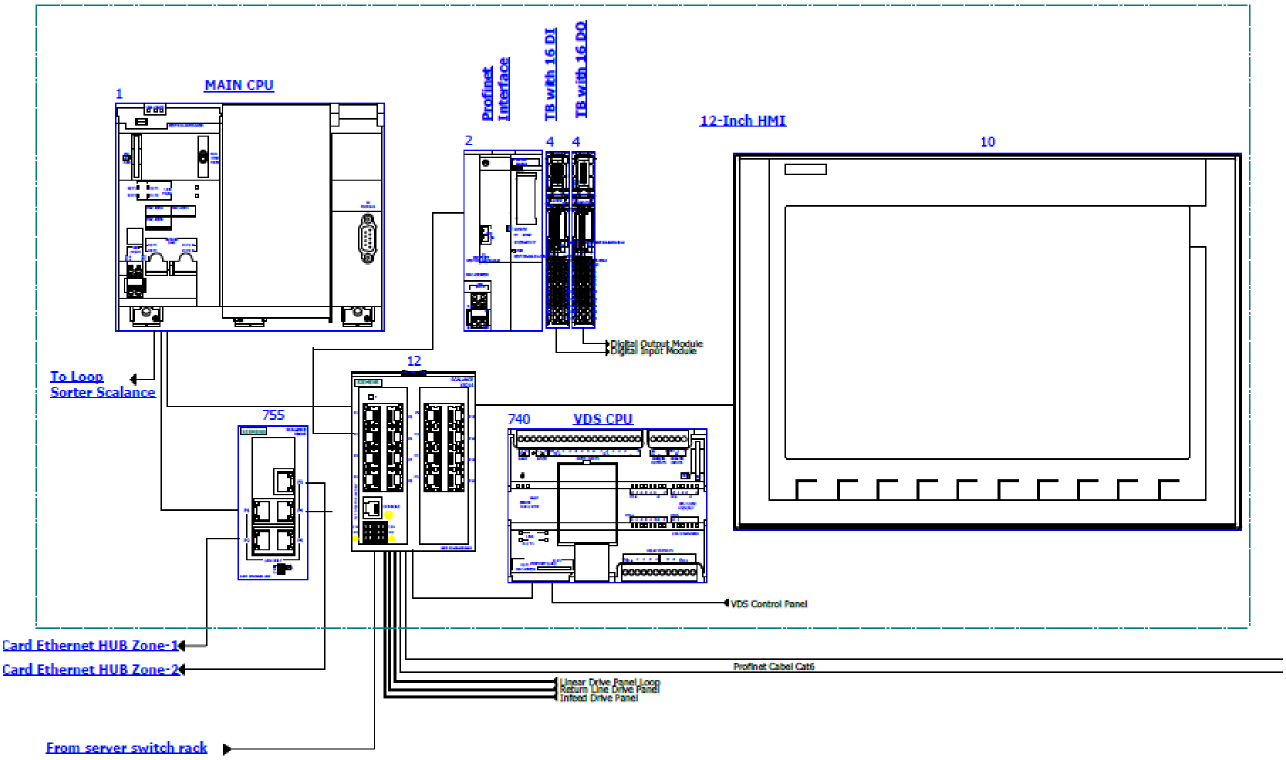
Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

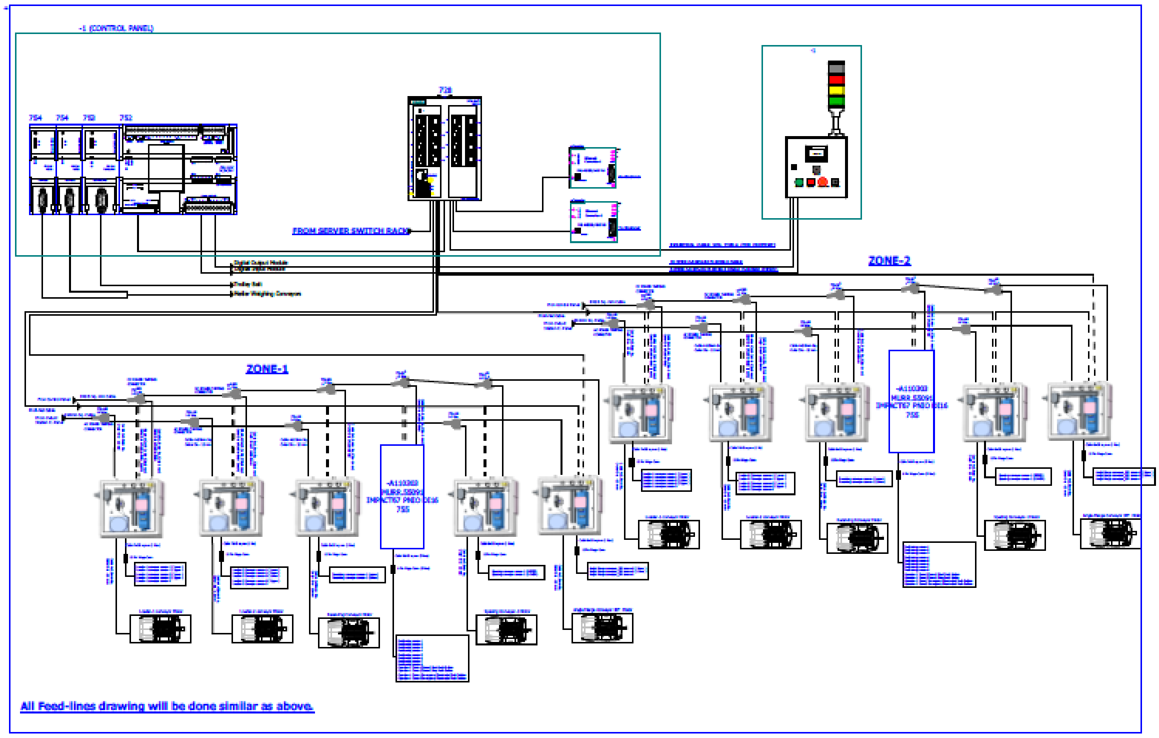
15.1 Reference Picture of Power Distribution Panel



15.2 Main Control Panel (Only for Reference)



15.3 Induct Stations Control Panel (Only for Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter.
The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.
All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

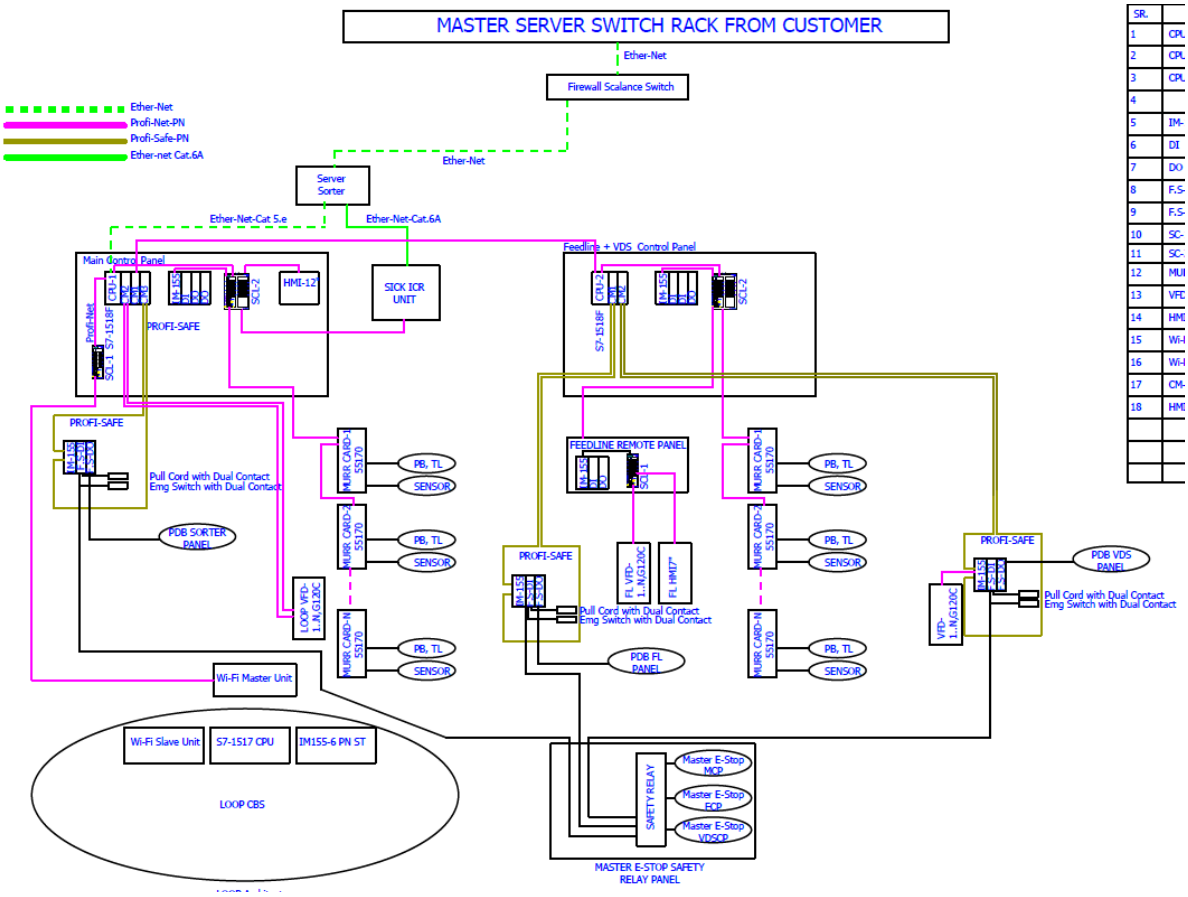
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Linear), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

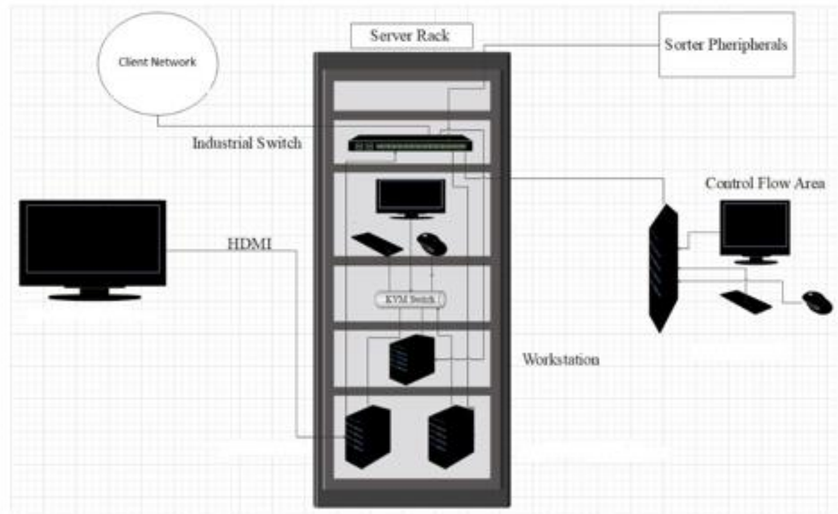
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

16. Controls Architecture (Only for Ref.)



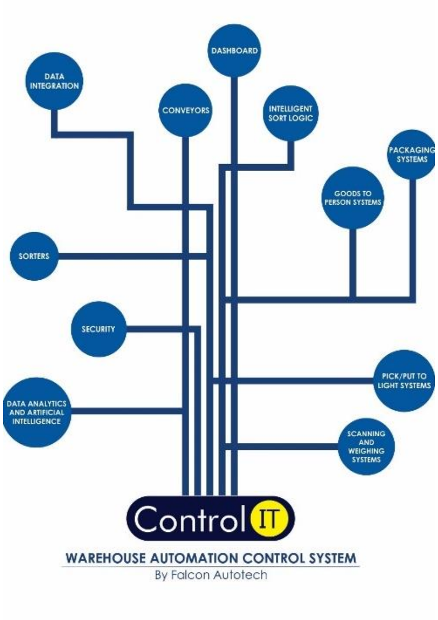
17. IT Architecture (Only for Ref.)



18.HAA Server for Sorter (Optional)

S.no	Description	Qty
	20Core Config with 128 GB RAM in T440 and 64 GB RAM in T40	
1	**Synology_storage_DS723+** 2.6 GHz AMD Ryzen R1600 Dual-Core, 2 x Gigabit Ethernet Ports,	2
	2GB ECC DDR4 RAM, 2 x 3.5/2.5" Bays | 2 x M.2 2280 Slots E10G22-T1-Mini 10GbE RJ-45 network upgrade module for compact Synology servers	
2	**Dell Tower Model T440** -PowerEdge T440 Xeon Gold 6148 2.4GHz/20C/27.5MB/150 W 16 DIMMS 4 x32GB RDIMM Up to 8 3.5"" Hot Plug Hard Drives Tower Configuration 2 x 1.2TB 10K RPM SAS 12Gbps 512n 2.5in	2
	Hot-plug Hard Drive PERC H750 Adapter Full Height 8GB Cache Dual Hot-plug PS495W iDRAC9Enterprise 3YR ProSupport Next Business Day Onsite/Dual Port Lan	
3	**Dell PowerEdge T40** Intel Xeon E-2224G Processor 3.5GHz 8M Cache,4C/4T,Turbo,71W,TPM, 4*16 GB RAM (4 DIMM), 2x1TB SATA 3.5" 7.2k rpm HDD (3 Bay), 1Gbe LOM ,DVD Writer, Onboard RAID 01, Inbuilt PSU Standard (Max 1), 3	1
	Year Onsite NBD,480 GB SSD,1 Gbe Dual Lan Card Extra	
4	Windows Server 2019	3
5	Monitor	1
6	KVM Switch	1
7	Mouse	1
8	Keyboard	1
9	Lan Cable	10
10	Power Cable	8
11	VGA Cable	1
12	42 AC Server Rack	1
13	Netgear 24 Port Giga Switch	2
14	2 TB SSD Micron	4
15	DP to VGA Convertor Cadyce Brand	1

19.Falcon's WCS CONTROLIT (For Reference Only)



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

20. Key Components Make

Items	Make
Belts	Forbo/ Derco
Rollers	Falcon
Cross Belt Carriers	Falcon
LSM	SEW
Feed Line Motors	Induction Motors – SEW/ Lenze/Similar
Scanners	Cognex
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Omron
Control Panels	Rittal/ BCH
VFDs	Siemens/Lenze/ AB/ Omron/Siemens
Cables	LAPP/ Equivalent
Switch Gear	Schneider/Equivalent
Bearings	NTN/SKF (Excluding Bearing of Rollers. Bearings will be FAPL /OEM standard)
Power Transmission Systems	Vahle
HMIs	Omron

21. XPRESSBEES's Responsibility

21.1 XPRESSBEES Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- XPRESSBEES to provide HPT/ Forklift for material movement for complete duration.
- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the operation of the shipmentsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment and makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation. This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.

21.2 Responsibilities of XPRESSBEES During the Tests

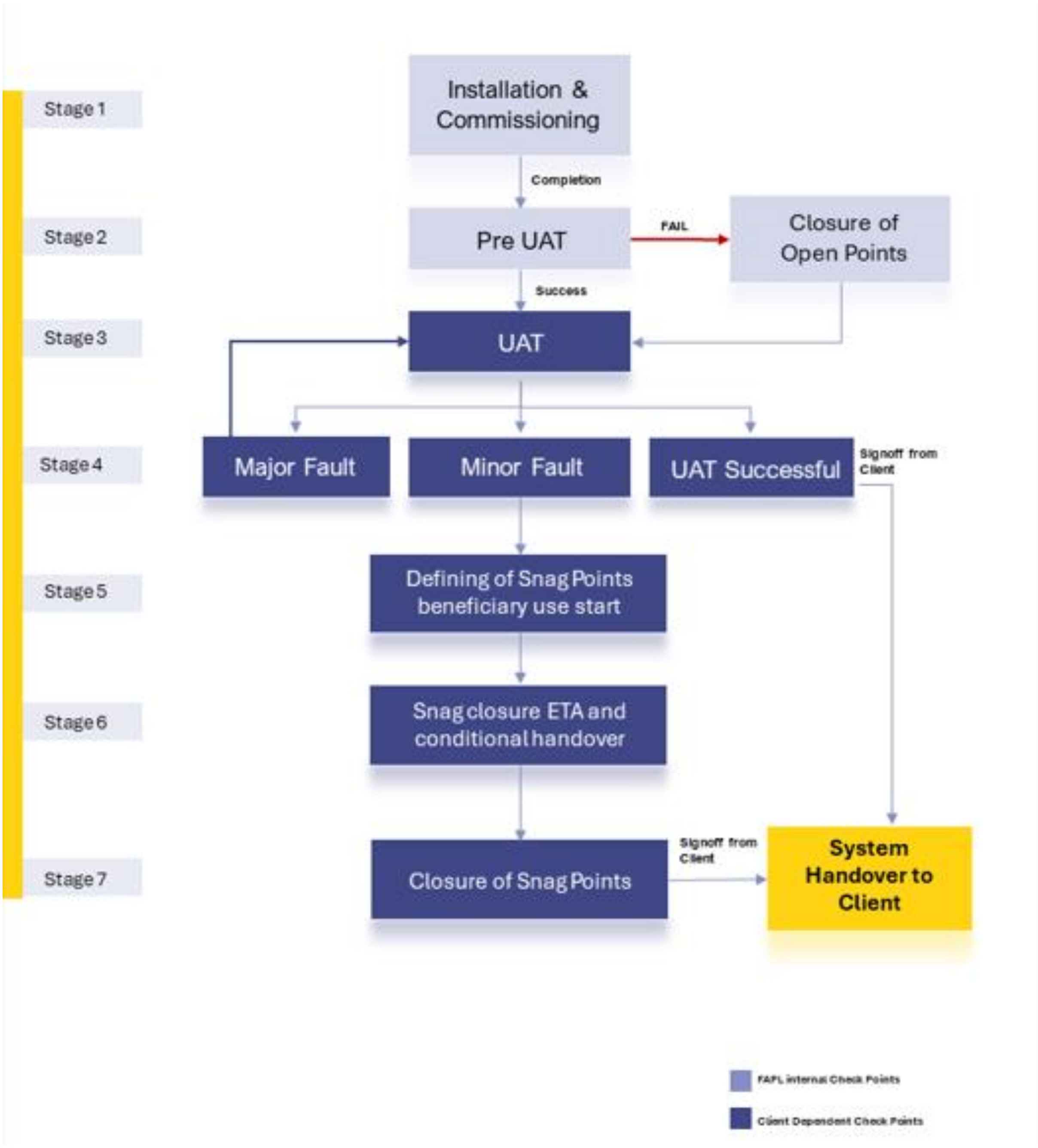
- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the shipments correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

21.3 XPRESSBEES Responsibilities During the Training

- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough spacefor desks or tables and chairs for the trainer and trained staff.
- Check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technicalstaff.
- The invitation of staff to attend the training courses will be at the expense of XPRESSBEES.

22.System Handover to XPRESSBEES

The system handover will be following the shown below workflow. Each of the stages are defined in the document below.



Installation and Commissioning – Completion of all the activities to get the system up and running.

Pre-UAT – Pre-UAT (User Acceptance Testing) involves a series of preparatory steps and tests conducted before the formal UAT phase begins. This stage ensures that the system is ready for end-user testing and meets the necessary requirements and standards. Below are some key checks of Pre UAT:

- **Physical Verification:** Ensure all functional and non-functional requirements are clearly defined and understood. Verify that all the necessary specifications for the sorting system are documented.
- **Integration Testing:** Conduct integration tests to ensure all components (conveyors, scanners, sorters, control systems, etc.) work together seamlessly. Validate the interfaces between the sorting system and other warehouse management systems (WMS) and databases.
- **Performance Testing:** Test the system's performance with some dummy product to ensure it can handle the expected volume of parcels or items. This test also done to identify and rectify any performance bottlenecks or inefficiencies.
- **Functional Testing:** Execute test cases to verify that the system performs all required functions correctly. Include scenarios for different types of products, error handling, and edge cases.

By thoroughly conducting pre-UAT activities, Falcon team will ensure that the system is stable, efficient, and ready for the formal UAT phase, where end-users can validate its functionality and performance in a real-world scenario.

UAT (User Acceptance Test)

User Acceptance Testing (UAT) for a Cross Belt Sorter System is the final phase in the implementation process where the end users test the system to ensure it meets their requirements and works as expected in a real-world scenario

Falcon will provide the UAT Schedule to Client prior to the commencement of any tests. The UAT Schedule will include the following:

- Test prerequisites.
- Daily plan/schedule
- Personnel responsibilities
- Dashboard data and test loads required.
- Specific test procedures
- Expected test results.

User Acceptance Test Parameters

Below are the parameters/test to be done at site during the UAT :

- Sorter Throughput Test
- Sorter Accuracy Test
- Barcode Reading Test

**SOP for all test will be shared during DAP stage. In case client require sample SOP for review , same will be shared during contract finalisation stage also.*

Note: In conducting this evaluation/UAT, Falcon team adhere to its Standard Operating Procedure

Minor & Major Faults

Any faults encountered during Testing will be categorized by Falcon as either Minor or Major faults as defined below.

● **Minor Faults** are defined as those affecting a limited area or single component that has no impact on the test. Minor Faults will be added to the System Snag point list and will not inhibit the continuation of testing.

● **Major Faults** are defined as those having impact that result in the inability to demonstrate the subject functionality. Major Faults may require re-starting of the test.

System Snag Point

Once the UAT completed, all the minor faults will be considered as snag points. Falcon team will publish Snag Point List information to the client.

System Snag Point list information will include the following:

1. Issue.
2. Date identified.
3. Area and/or unit.
4. Category (mechanical, electrical, or software).
5. Remedy responsibility (responsible person[s] or organization[s]).
6. Target completion date.
7. Snag point completion verification and signoff.

Once the snag point list information's shared with client, client can start the beneficiary use of system and also Client needs to sign off on Start of System Warranty and conditional handover .

Practice for Snag Point Sign Off:

1. Falcon personnel will maintain and distribute the System Sang List throughout UAT.

2. As each issue is corrected, Client will verify the resolution and provide sign-off for that issue on the System Snag Point List.

System Handover Letter

After successful completion of Acceptance Testing or closure of all snags, Falcon will furnish Client a letter, which will address the following:

- The subject System has been installed and accepted by Client.
- The final payment amount that is required and the designated due date.

23.Commercial Terms

23.1 Price Sheet

For Option 1- With 8 Destinations

S. No	Component	Qty/Set	Set	UOM	Price
1	Sorter Package	1	1	Set	₹ 73,64,827
2	Conveyor Package	1	1	Set	₹ 12,24,788
3	Installation & Commissioning	1	1	Set	₹ 4,29,481
4	Packaging & Documentation	1	1	Set	₹ 1,71,792
Total					**₹ 91,90,888**

For Option2- With 40 Destinations

S. No	Component	Qty/set	Set	UOM	Price
1	Sorter Package	1	1	Set	₹ 1,70,02,565
2	Conveyor Package	1	1	Set	₹ 12,24,788
3	Installation & Commissioning	1	1	Set	₹ 9,11,368
4	Packaging & Documentation	1	1	Set	₹ 3,64,547
Total					**₹ 1,95,03,268**

Optional Items

S. No	Component	Qty/set	Set	UOM	Price
1	HAA Server for Sorter	1	1	Set	₹ 15,89,230
2	Mezzanine/Operator Platform	1	1	Set	₹ 17,08,612
3	Sort Fail Conveyor System	1	1	Set	₹ 3,67,925

23.2 Pricing Terms:

- Prices are Ex works
- Scope as described in the offer – Included
- Taxes extra as applicable
- Freight & Unloading at Site: Client Scope
- Laydown area & material security at site are in customer scope
- Price is Valid for 30 Days

23.3Payment Terms

Payment Percentage	Stage
40%	Advance along with PO
50%	Against Delivery with 100% Taxes
10%	Against Installation

24.Warranty Period

Falcon offers Comprehensive warranty for 1 year (Starts from the date of beneficiary use, max 30 days after readiness of commissioning).

The warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in the Exclusion Clause)

The warranty does not apply to the replacement or repair of:

- Normal wear and tears.
- Consumables.
- Faulty articles continued:
    - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
    - Negligence or abnormal use of equipment.
    - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
    - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that is supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

Note: Transportation cost for sending faulty items to Falcon Factory from the site will be in XPRESSBEES Scope and new items from Falcon factory to site will be in scope of Falcon.

25. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Construction Power
- Maintenance Platform / Lift required for maintenance activity
- Collection Bins / roller cages / pallets
- Safety Fencing
- SCADA
- Workstations
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices.
- Traffic and route markings
- Laydown Area
- Ram protection devices / crash barriers
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.

26. Appendices

1. Solution Layout: LCBS with 8 Destinations *
2. Solution Layout: LCBS with 40 Destinations *

* A free Navis viewer can be downloaded here: https://www.autodesk.com/products/navisworks/3d-viewers